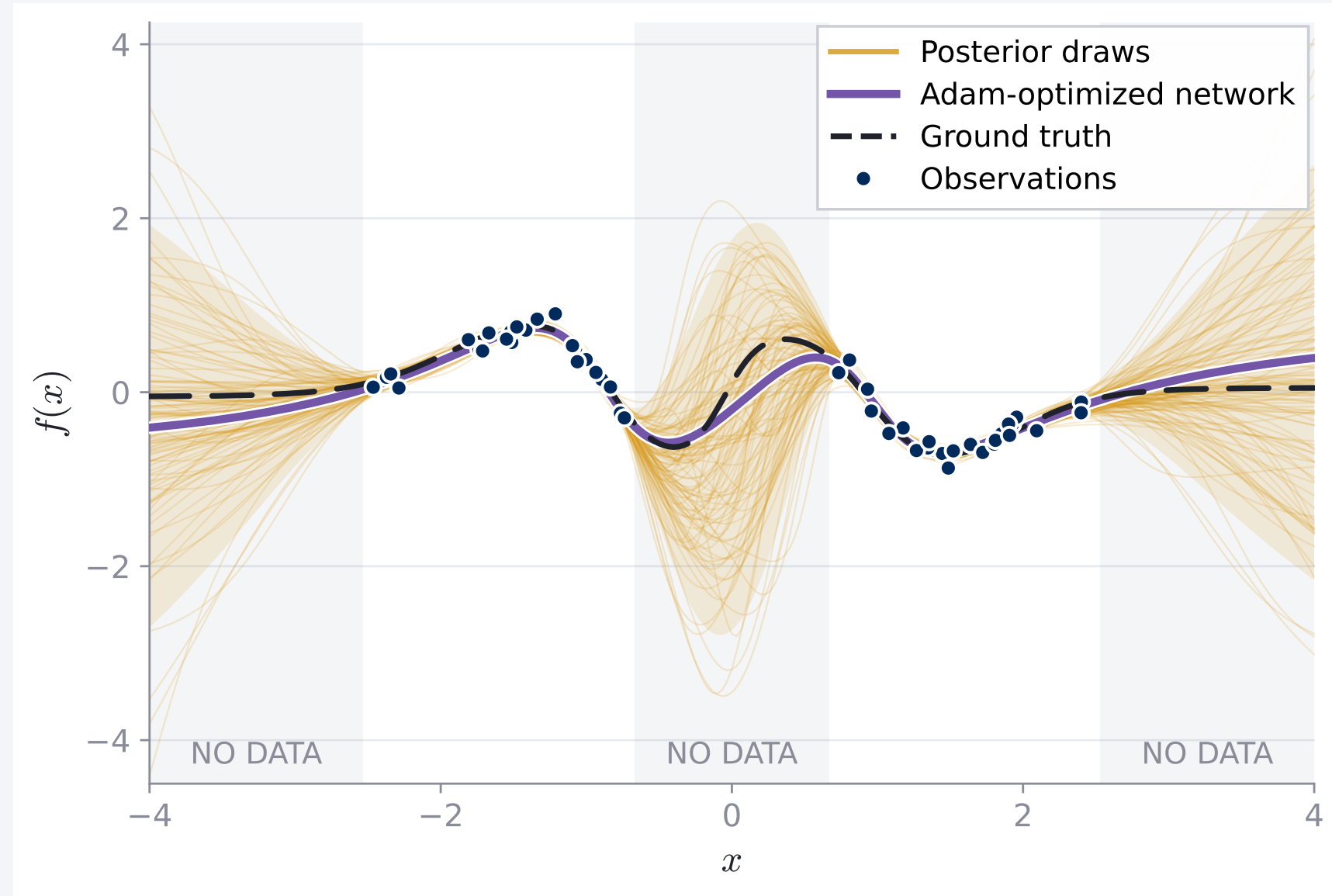


Why sample a neural network?



Ordinary training: one set of weights $\rightarrow$ one function.

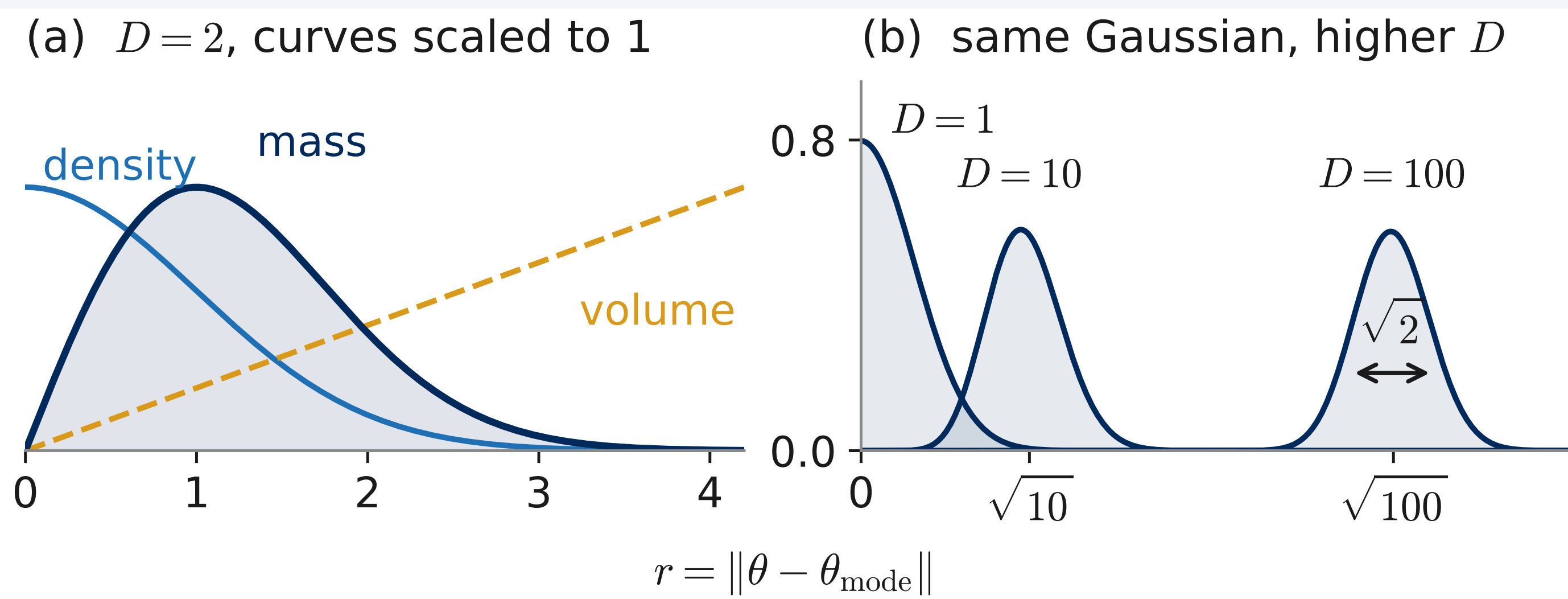
Bayesian sampling: a distribution over weights $\rightarrow$ uncertainty over functions.

$$\pi(w) = p(w | \mathcal{D}) \propto \underbrace{p(\mathcal{D} | w)}_{\text{fit}} \underbrace{p(w)}_{\text{prior}}$$

Every gold curve fits the data we have. They only disagree out in the shaded gaps, and that disagreement is the uncertainty we are after.

Where the probability lives

$$\mathbb{E}[f] = \int f(\theta) p(\theta) d\theta = \int f(r) p(r) r^{D-1} dr$$



Density peaks at the mode. **Volume** grows as r^{D-1} , for D parameters. Their product is the *typical set*: a thin shell at $\|\theta\|^2 \approx D$, relative width $\sqrt{2/D}$. A sampler has to stay on that shell. If something quietly adds energy the chain drifts outward, and none of the usual convergence diagnostics complain.

MCMC in the era of big data

You cannot draw from π directly. A chain can end up there:

$$w^{(0)} \rightarrow w^{(1)} \rightarrow \dots \rightarrow w^{(M)}, \quad \{w^{(m)}\}_{m \gg 0} \sim \pi.$$

Gradient samplers steer by $\nabla \ln \pi$; Hamiltonian Monte Carlo (HMC) is the standard one. Computed exactly, every step reads the whole dataset:

$$\nabla \ln \pi(w) = \sum_{i=1}^N \nabla \ln p(y_i | w) + \nabla \ln p(w).$$

A minibatch $B \ll N$ is the standard fix:

$$\hat{g}_B = \frac{1}{B} \sum_{i \in B} \nabla \ln p(y_i | w) + \nabla \ln p(w),$$

$$\mathbb{E}[\hat{g}_B] = \nabla \ln \pi(w), \quad \text{Cov}[\hat{g}_B] \propto 1/B.$$

The mini-batch gradient is unbiased, but no individual step ever uses the right one. How much that per-step error costs you is what we set out to measure.

Why ray tracing tolerates noisy gradients

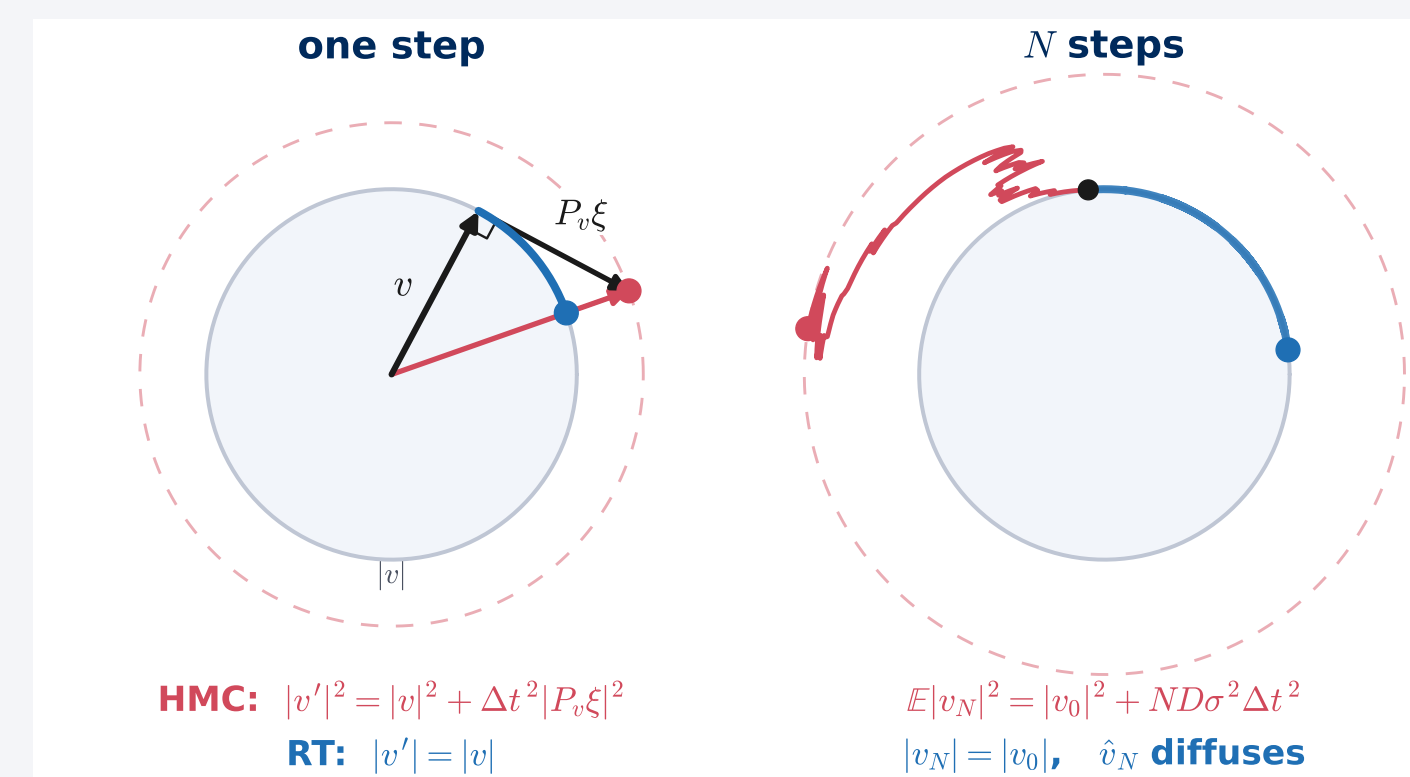
1. Noise speeds HMC up. It only rotates a ray.

HMC adds the gradient onto its velocity. A random push lengthens that velocity whichever way it points, and nothing takes the energy back: the chain heats up, drifts *outward* off the shell, and its error bars come out too wide.

Ray tracing treats the posterior as an optical medium. Setting the refractive index to $n(\theta) = \pi(\theta)^{1/(D-1)}$ makes rays bend toward higher probability. The sampler carries a *unit* direction of travel u , and the gradient error enters as $\sigma \zeta$:

$$\frac{du}{ds} = P_u \left(\nabla \ln n(\theta) + \frac{\sigma}{D-1} \zeta \right), \quad P_u = I - uu^\top.$$

Both act only in the tangent plane, so they can only rotate u on the constant-speed sphere.



2. Many noisy turns become angular diffusion.

Write $b(h, \sigma)$ for the bias the noise leaves in equilibrium, and $\sigma_c(h) \sim h^{-p}$ for the noise tolerated at step h . Averaged over the noise, each kick contributes $\sigma^2 h^2 / 2(D-1)^2$ of diffusion Δ_S across the sphere of directions, and a trajectory of length S holds S/h of them:

$$\frac{S}{h} \cdot \frac{\sigma^2 h^2}{2(D-1)^2} \Delta_S = O(\sigma^2 h) \Delta_S,$$

the order in h that leaves HMC at $p = \frac{1}{2}$.

3. The leading diffusion preserves equilibrium.

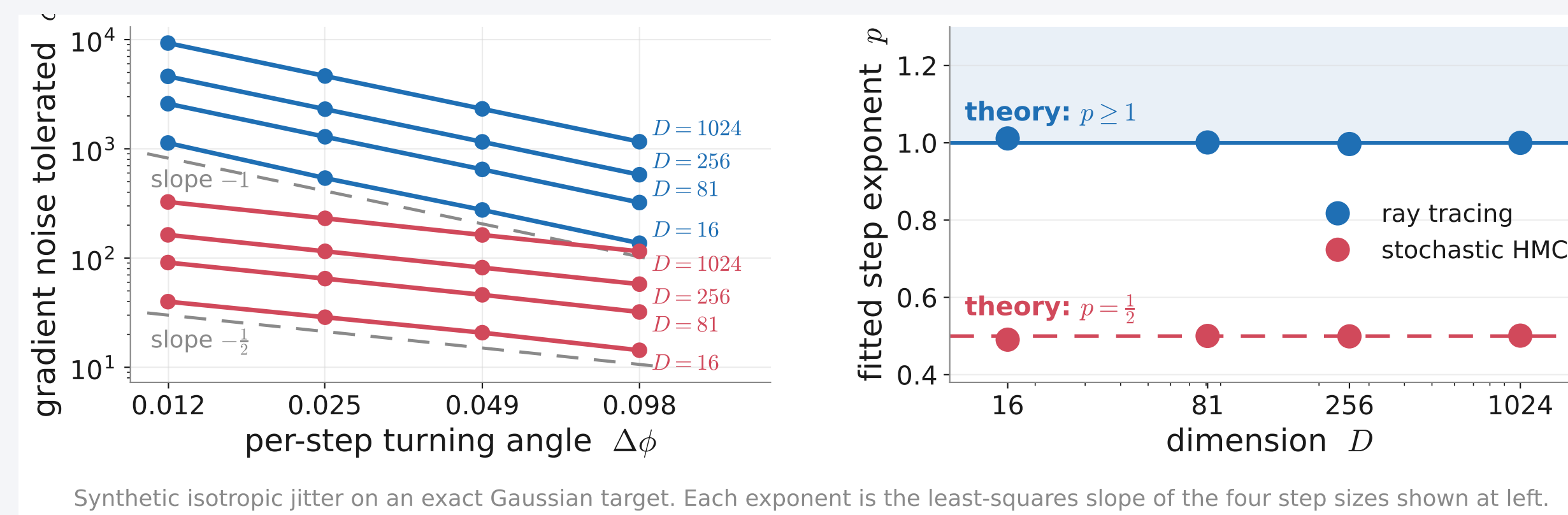
$$\begin{aligned} \rho_0(\theta, u) &= \pi(\theta) \mu_S(u), & \Delta_S \rho_0 &= 0 \\ \implies b(h, \sigma) &= O(\sigma^2 h^2) + o(\sigma^2) \\ \implies \sigma_c(h) &= \Omega(h^{-1}), & p &\geq 1 \end{aligned}$$

The cancellation is $\Delta_S \rho_0 = 0$: diffusion cannot spread directions that are already uniform, and clean RT transport preserves ρ_0 too.

Individual paths wander at $O(\sigma^2 h)$. The distribution they settle into does not move at that order.

4. Is the bound tight?

Theory gives $p \geq 1$ for ray tracing against $p = \frac{1}{2}$ for HMC, so halving the step buys HMC 1.4 $\times$ the tolerable noise and ray tracing at least 2 $\times$.

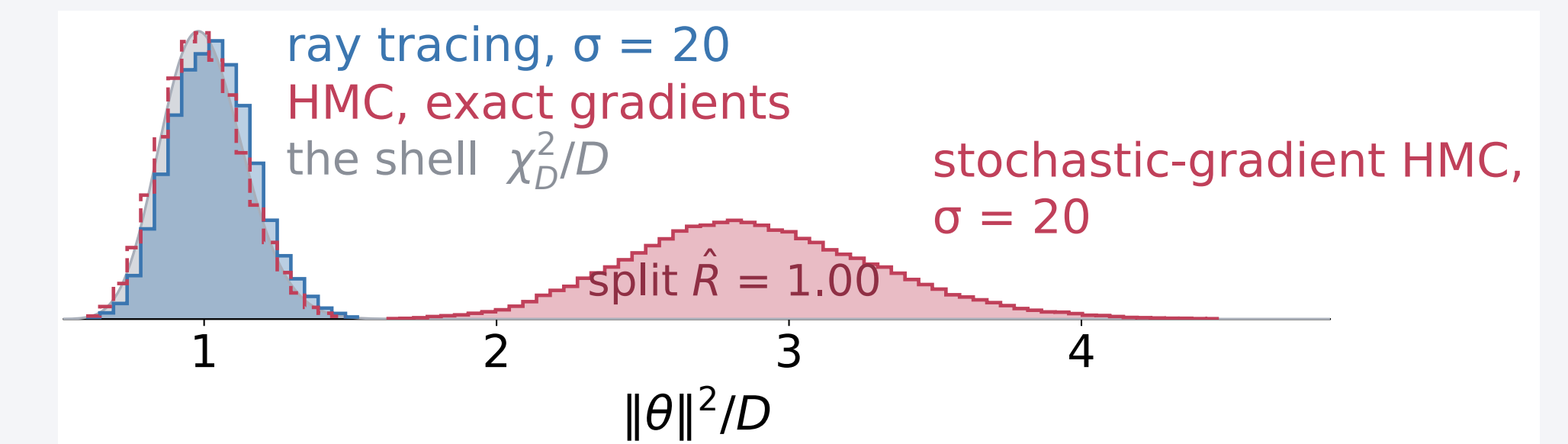


The bias is read against the clean chain at the *same* step, so integration error cancels:

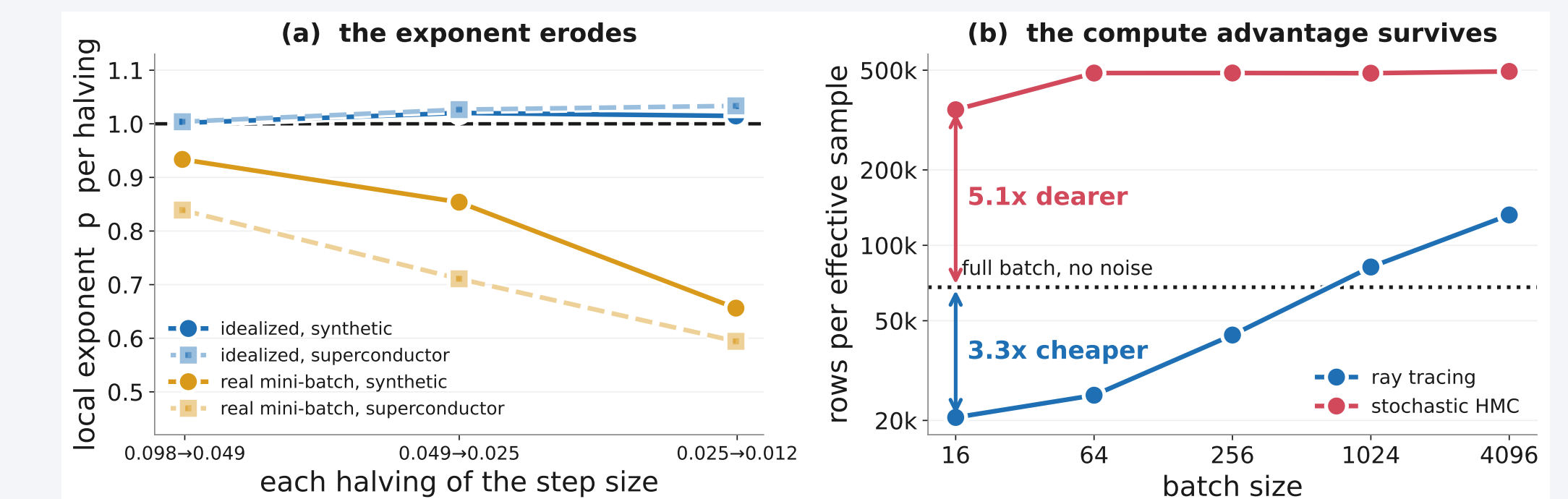
$$b = [\mathbb{E}_{h,\sigma} \|\Theta\|^2 - \mathbb{E}_{h,0} \|\Theta\|^2] / D = (\sigma/\sigma_c)^2 + o(\sigma^2).$$

Failure modes, measured

Injected noise, $\sigma = 20$, $D = 100$: stochastic-gradient HMC equilibrates three times off the shell with split $\hat{R} = 1.00$, no alarm; ray tracing holds the shell.

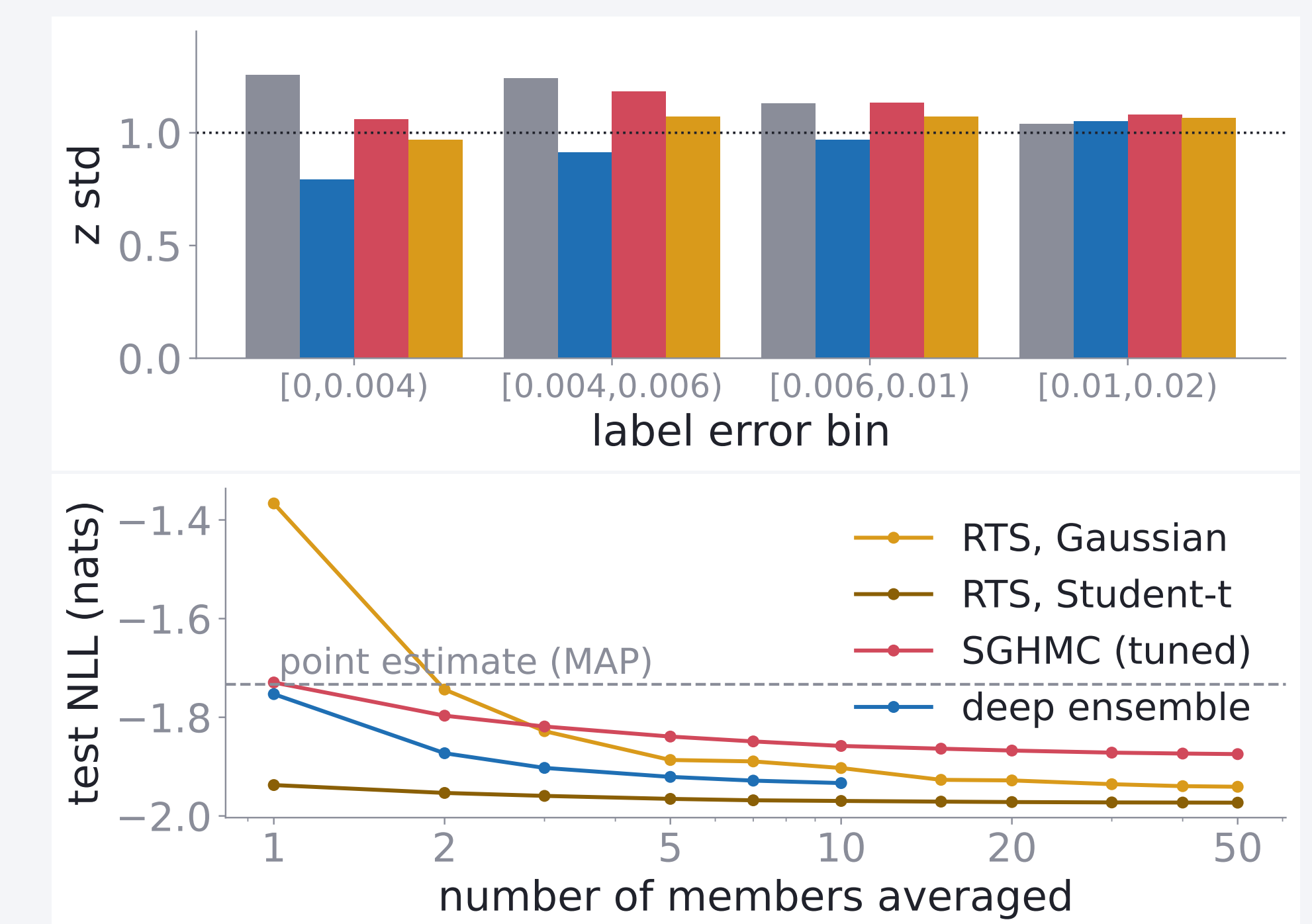


The theory assumes the gradient error is independent of position and symmetric in sign. Force both onto a real minibatch (*idealized* below) and the exponent holds; leave the batch as it comes and it decays with every halving, so no single p survives.



Uncertainty quality on the Gaia posterior

Gaia XP regression, 126,156 stars, $D = 11,394$, sampled with minibatch ray tracing. Key: **point (MAP)**, **deep ensemble**, **SGHMC**, **ray tracing**.



Conclusion

- Gradient noise can only rotate a unit ray, and the fitted step exponent confirms the bound: $p = 1$ at every dimension tested ($D = 16$ to 1024), against $p = \frac{1}{2}$ for stochastic HMC.
- Real minibatches break the noise assumptions and erode the clean exponent, yet the compute advantage survives (3.3 $\times$ cheaper per effective sample; stochastic HMC 5.1 $\times$ dearer). Still open: whether $p = 1$ is exact in stationarity.
- On a real survey posterior, only ray tracing keeps per-group calibration flat, and member averaging passes a deep ensemble on test NLL.

Acknowledgments

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